Probability Distribution

Pooja Panwar[1]
Anant Deshwal[2]

[1]University of Jyväskylä
[2]Bradley University

Credit to:

- Tom Hobbs (Colorado State University)
- Mevin Hooten (University of Texas)
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Probability Distribution and Random Variable

- Equation of line:
    - **y= mx+b** *or* **y = *f(x)*** where for m and b help map x to a single value of y or *f(x)*.
- **Probability Distribution** tells us how possible values of a random variable are distributed and their chances of occuring.
    - **Random Variable**: A function that represents outcome of a random process using values on real line. *Here y and x are both R.V.*
        - Roman Lettes (X, Y, etc.) - for Random Variables
        - Greek Letters (α, γ, θ, etc.) for population parameters

Examples

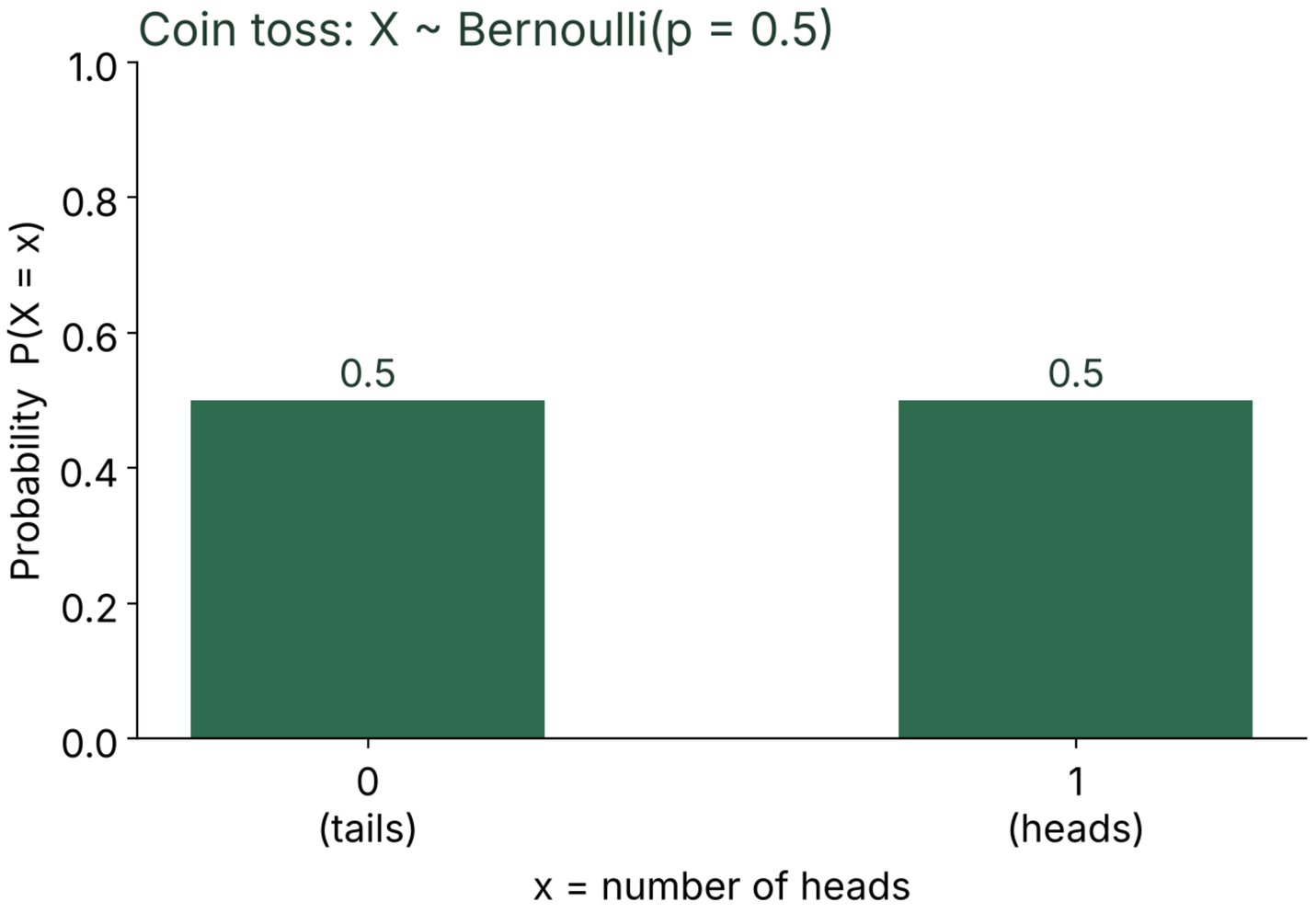
Random Variable: A function that represents outcome of a random process using values on real line.

- Example 1:
  - Random Process: Coin Toss
  - Possible Outcomes: Heads or Tails
  - Random Variable: X = number of heads (or can be tails)

- Example 2:
  - Random Process: Number of Species recorded in an Urban Habitat
  - Possible Outcomes: Any whole number (0,1,2,3,….,n)
  - Random Variable: Y = number of species

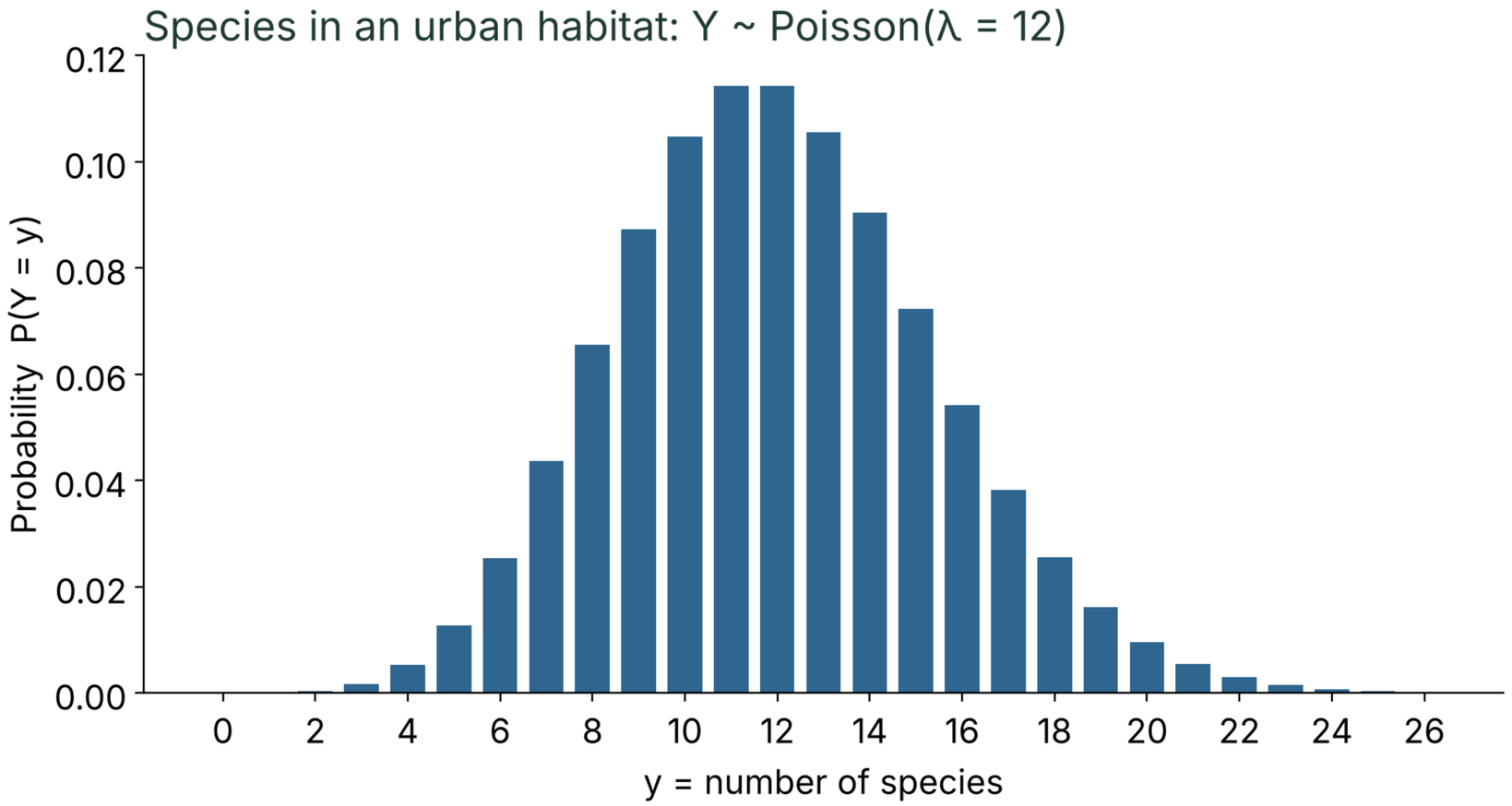
What would the data look like?

Example 1:



What would the data look like?

Example 2:

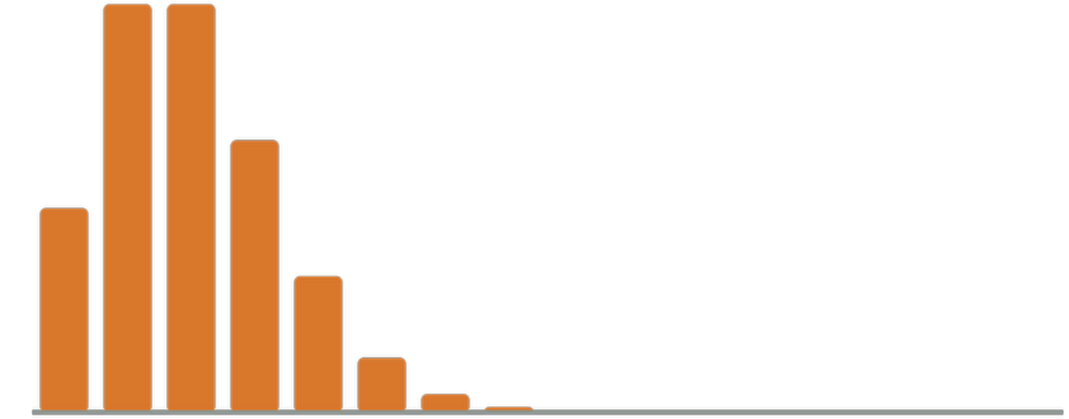


Types of Probability Distributions

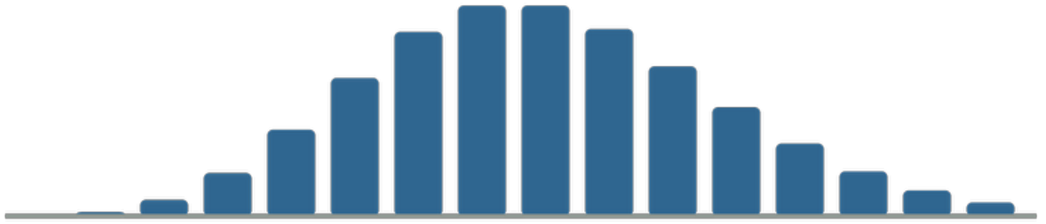
- Normal Distribution
- Poisson Distribution
- Binomial Distribution
- Bernoulli Distribution
- Negative Binomial Distribution
- Gamma Distribution
- Beta Distribution

Poisson distribution

- Mean = Variance
- Parameter: ƛ
- Takes on whole number
  - Species Richness
  - Population Size



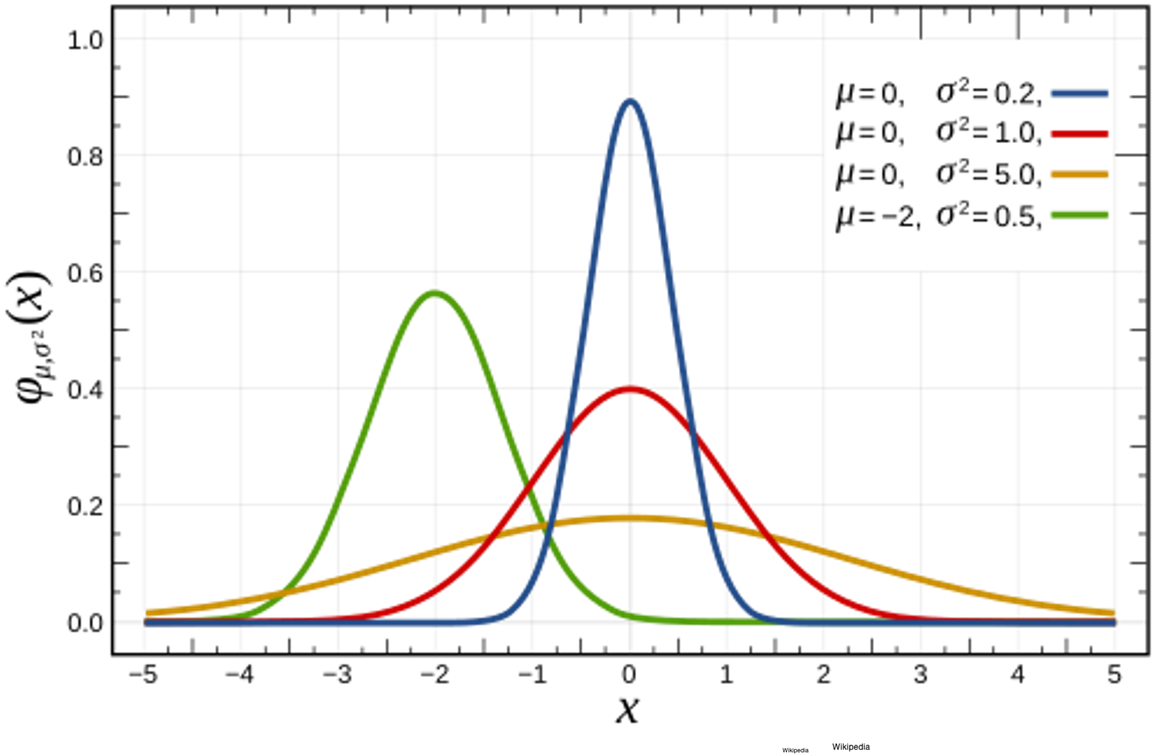
λ = 2: a dry, open park



λ = 8: a park with deep litter

Normal Distribution

- Mean = Median= Mode (Symmteric around Mean)
- Parameters: σ^2 and μ
- Continuously distributed, can take +ve and -ve values

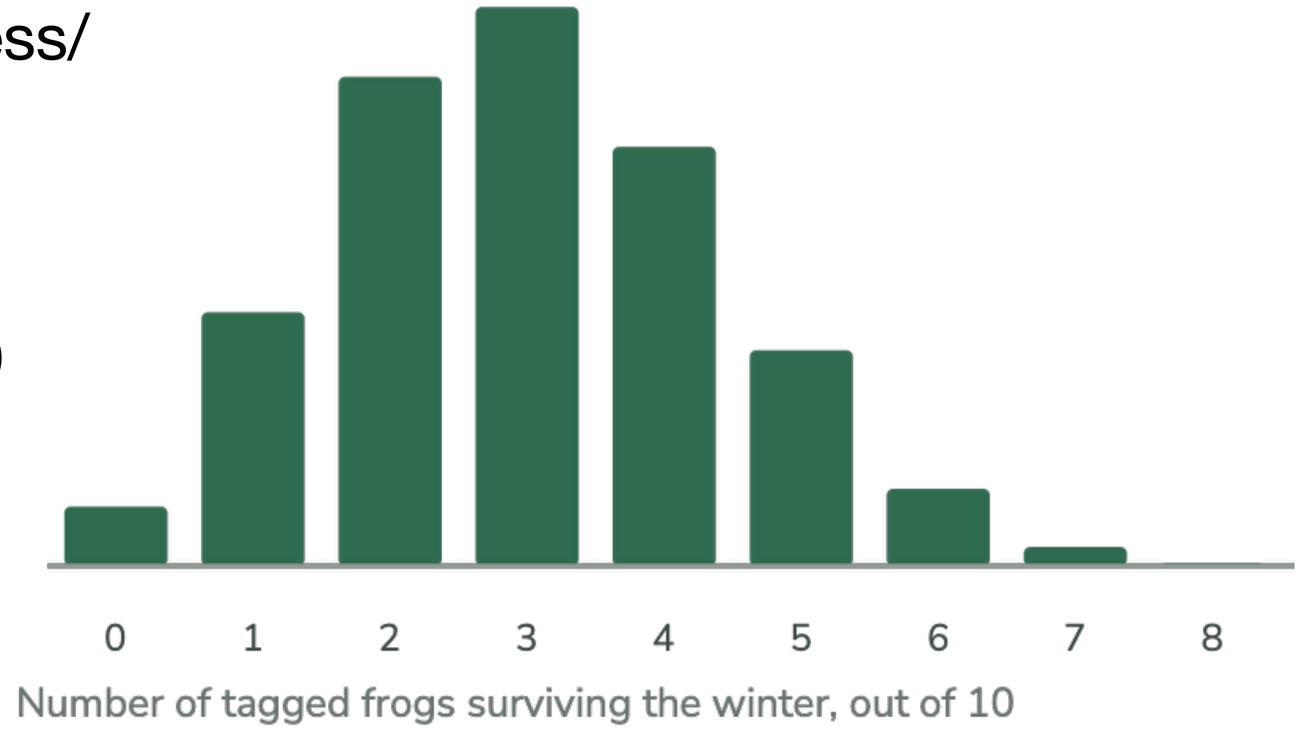


Wikipedia Wikipedia

Binomial Distribution

- Number of success in a fixed number of independent trials
- Only two possible outcomes: Success/ Failure. Survival/Death. Heads/Tails
- Parameters: number of tries (n) and success of probability in each try (p)

Number of tries: 10 (fixed)
Outcomes: Frog Survives or Dies
Same probability of survival for each frog: 0.3
N cannot be greater than 10



Bernoulli – single trial with 2 outcomes (success/failure) (yes/no)(0/1)

Is the invasive species present in this plot? Did this one frog survive?

Negative Binomial – Number of trials needed to achieve specified number of success with success probability p

Ecology: Overdisepersed Poisson. Mean < Variance.

K - dispersion parameter.

Uniform - Has upper and Lower values. Every value in range equally likely.

Often used as prior for σ

Gamma – positive continuous distribution. Right Skewed.

Bodymass, amount of rainfall, waiting time between events, areas, distances

Beta – continuous distributions from 0 to 1

Cheatsheet

Distribution	Use it for	Parameters	In JAGS	Ecology example
Normal	Measurements	μ, σ	dnorm(mu, tau)	Litter depth, body mass
Uniform	Any value in a range	lower, upper	dunif(0, 100)	Prior for σ
Poisson	Unbounded counts	λ	dpois(lambda)	Beetles per trap
Binomial	Successes out of n	p, n	dbin(p, n)	Frogs surviving of 10

Help me!

- What Distribution would you choose?
  - Number of earthworms in a soil core
  - Mass of Golden-crowned kinglets
  - 200 seeds planted, seeds germinating. Each seed has equal probability of germination
  - Percentage Urbanization